

Chang-Hao Yang

✉ kenny103089@gmail.com
☎ 0919430056

🐙 github.com/Kenny0711
in in/YangChanghao0711

EDUCATION

National Yang Ming Chiao Tung University
M.S. in Computer Science

Taiwan, Hsinchu
Nov 2025 – Present

Yuan Ze University
B.S. in Computer Science

Taiwan, Taoyuan
Nov 2020 – Jun 2024

- Relevant Coursework: Operating Systems, 3D Game Design, Algorithms

ADVANCED TECHNICAL SKILLS & KNOWLEDGE AREAS

- **Languages:** Python, C, C++, C#, Java
- **Frameworks & Libraries:** PyTorch, NumPy, OpenCV
- **Tools:** Git, Docker

INTERNSHIP EXPERIENCE

Intern, Networking Product Testing

Jan 2023 – Jan 2024

- Tested AP, AC, and CC (Nebula) integration in different networking scenarios.
- Conducted stability, performance, and compatibility testing.
- Simulated network conditions and evaluated AP throughput in the shielding room.

PROJECTS

Neural Network from Scratch

2026

- Built a neural network from scratch using Python standard libraries and NumPy without external deep learning frameworks.
- Implemented forward propagation from the input layer through hidden layers to the output layer for prediction.
- Used backpropagation with the chain rule to compute gradients and update weights.
- Evaluated the model on XOR and linear datasets, and compared performance under different activation functions and optimizers.

Unity 3D Game

2025 – 2026

- Developed a 3D game in Unity using C# for a 3D game design course.
- Implemented game logic, player interaction, and scene-based mechanics.
- Designed three playable levels with different layouts and progression challenges.
- Demo video: [YouTube Link](#)

MaskGIT for Image Inpainting

2025 – 2026

- Implemented a Bidirectional Transformer-based MaskGIT model for image inpainting using PyTorch, leveraging a pretrained VQGAN encoder and codebook for discrete visual token representation.
- Built a custom Multi-head Self-Attention module with scaled dot-product attention and Gumbel-Max sampling for stochastic inference.
- Designed an iterative masked decoding pipeline (MVTM) with confidence-based token selection and support for cosine, linear, and square mask scheduling strategies.
- Achieved a best FID score of 28.24 using cosine scheduling with 10 inference iterations on a cat face dataset.